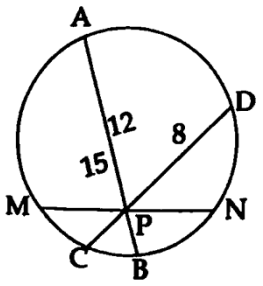
CIRCLES AND LINES

PYQ • 2021 - 2026

1. 2025 **5 MARKS**

In the figure, the chords AB, CD and MN intersect at P. PA = 12 centimetres, AB = 15 centimetres, PD = 8 centimetres.



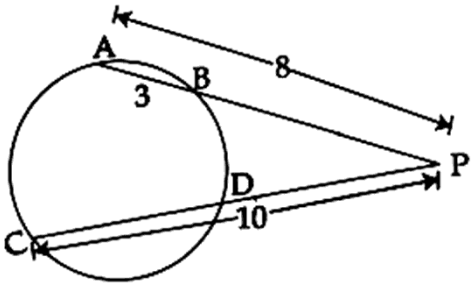
(a) Find the length of PC.

(b) Find the length of CD.

(c) If PM = PN, find the length of MN.

2. 2025 **3 MARKS**

In the figure PA = 8 centimetres, AB = 3 centimetres, PC = 10 centimetres.

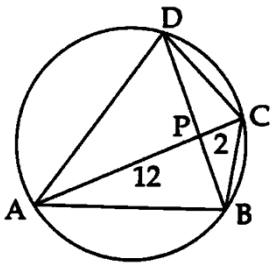


(a) Find the length of PB.

(b) Find the length of PD.

3. 2023 **4 MARKS**

Diagonals AC and BD of the cyclic quadrilateral ABCD cuts at P. PA = 12 centimetres; PC = 2 centimetres; BD = 11 centimetres.



(a) If PB = x, then write PD in terms of x.

(b) Find the lengths of PB and PD.

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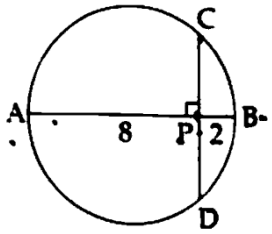
4. 2022 **2 MARKS**

In the figure AB is the diameter of the circle. Line CD is perpendicular to AB. AP = 8 centimetres and PB = 2 centimetres. Find the length of PC.



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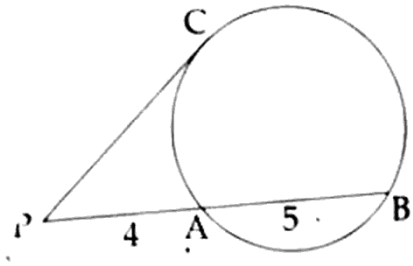
5. 2022 **2 MARKS**

In the figure PA = 4 centimetres, AB = 5 centimetres and PC is a tangent to the circle. Find the length of PC.



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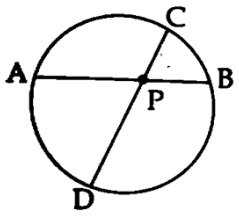
6. 2021 **3 MARKS**

In the figure chords AB and CD intersect at P. AB = 10 centimetres, PB = 4 centimetres and PC = 3 centimetres.



(a) What is the length of PA ? **(1)**

(b) Find the length of PD. **(2)**